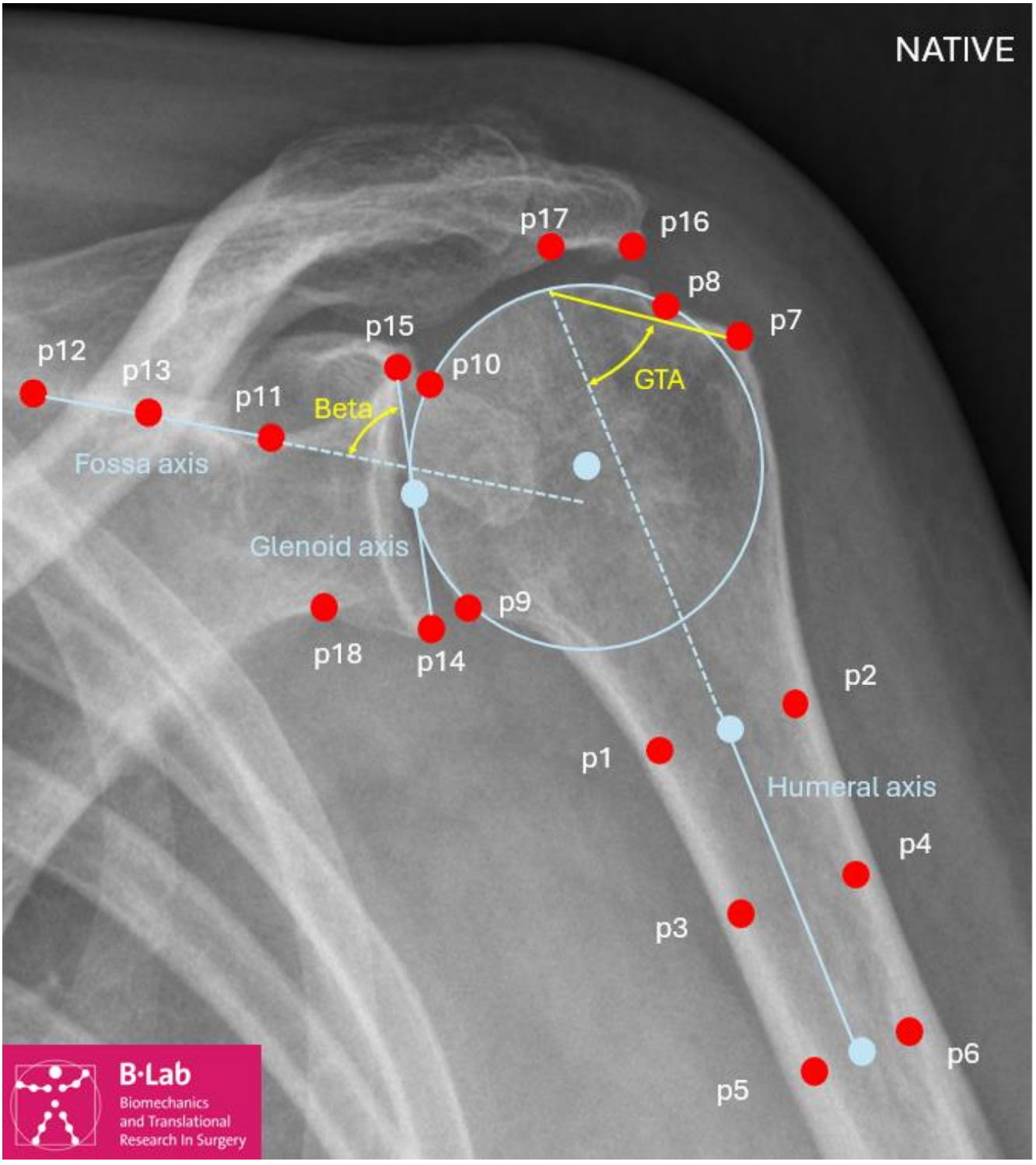
Point Placement Guide
Shoulder Analysis (Beta_angle_ana)

As a reminder, this extension enables the computation of shoulder morphological measurements on radiographs (or 2D images) based on precise anatomical landmarks. The accuracy of the results directly depends on the precision of point placement. Following, you will find a detailed description to help you place the fiducials nodes.



P1–P6 — Humeral diaphysis (proximal/distal, medial/lateral) — inner cortex
Place six points along the external cortical borders of the humeral shaft, in this order: proximal medial (P1) → proximal lateral (P2) → distal medial (P3) → distal lateral (P4). P1-P3-P5 form the medial cortical border and P2-P4-P6 form the lateral cortical border. The humeral shaft axis is computed as the least-squares centerline running midway between these two lines.

P7 — Latero-superior greater tuberosity (GT)
Place a point at the latero-superior edge of the greater tuberosity.

P8 — Upper head / greater tuberosity junction
Place a point on the upper aspect of the humeral head, at its junction with the greater tuberosity. Together with P7 and P8, this point is used to reconstruct the circle of the humeral head.

P9 — Lower head-calcar junction
Place a point at the lower head-calcar junction.

P10 — Head surface
Place a point roughly midway between P6 and P7 on the humeral head contour. Together, P8, P9 and P10 define the circle used to estimate the center of rotation of the humeral head — space them as widely as possible along the contour for a more accurate fit.

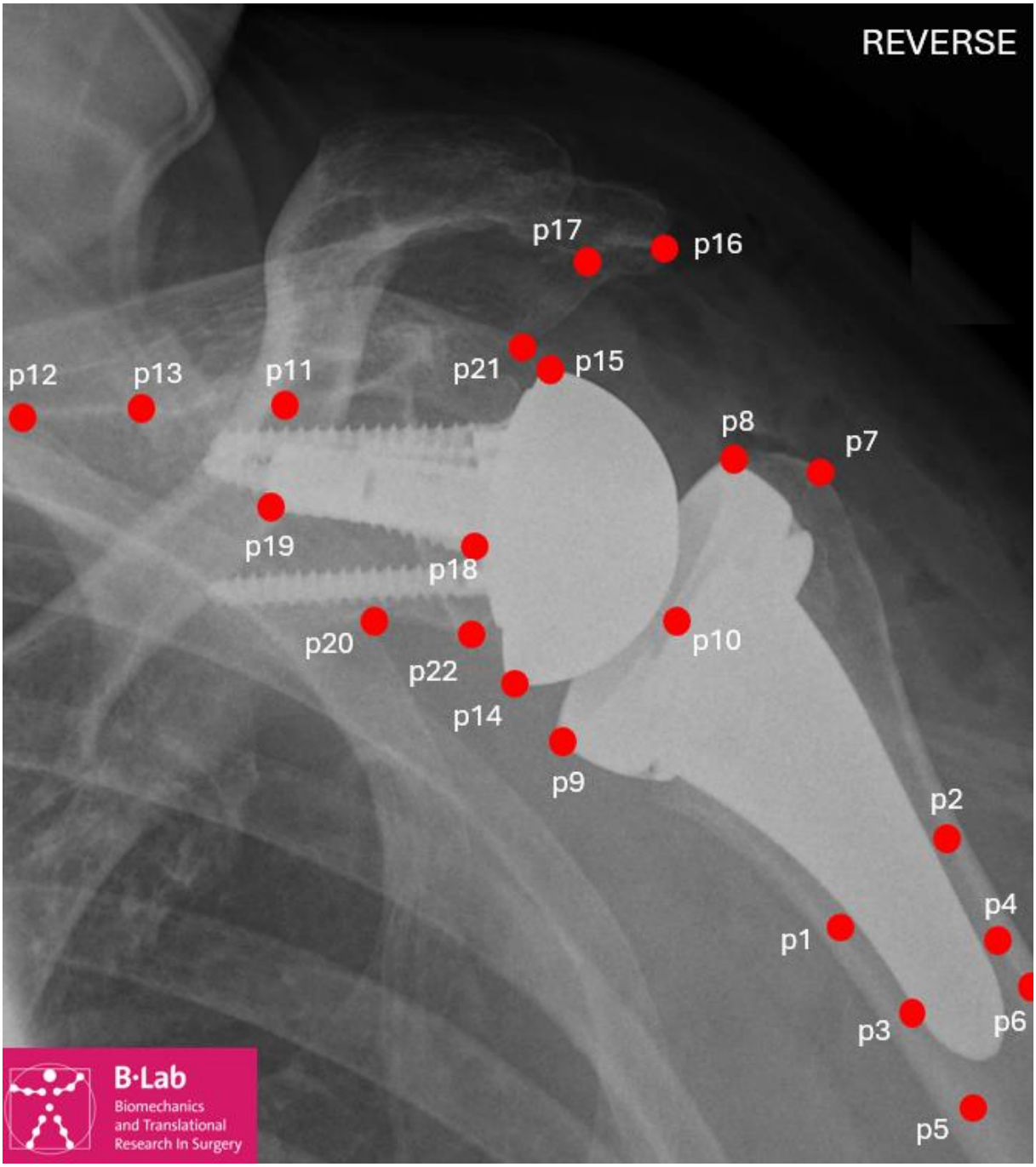
P11, P12, P13 — Fossa axis
P9: place at the "bottleneck" where the coracoid process broadens away from the scapular fossa.

P10: place at the most medial point of the fossa visible on the image.
P11: place on the fossa cortex roughly midway between P9 and P10 — this third point allows the fossa axis to be refined as a least-squares line fit rather than a simple two-point line. Together, P9, P10 and P11 define the fossa axis, used as the scapular reference line.

P14, P15 — Lower glenoid / Upper glenoid
Place points at the most superior and most inferior aspects of the glenoid rim. These define the glenoid line.

P16, P17 — Lateral acromion / Medial acromion (inferior border)
Place two points at the lateral and medial edges of the inferior acromial border. These define the acromial line.

P18 — Medial scapular neck
A line, parallelled to the fossa axis and passing through p12, will appear to guide the placement of p16. Indeed, the point should be placed where this line crosses the cortical edge of the scapular neck, to represent the base of the scapular neck.



For the reverse prosthesis the landmarks should be placed as shown above. Points P1 to P7 do not change.

P8 — Upper limit of the humeral component

Place a point on at the edge of the upper aspect of the humeral component of the reverse prosthesis.

P9 — Lower limit of the humeral component

Place a point on at the edge of the lower aspect of the humeral component of the reverse prosthesis.

P10 — Humeral component surface
Place a point roughly midway between P6 and P7 on the humeral component anterior edge.

P11-P12-P13 do not change

P14, P15 — Lower glenoid / Upper glenoid
Place points at the most superior and most inferior aspects at the base of the glenoid sphere.

P16-P17 do not change

[New] P18, P19 – Peg start and peg end (inferior edge)

Place both points along the inferior edge of the pegged glenoid component, on the line of the peg's anchor.

- P18 marks the start of the peg, at the limit between the glenoid sphere and the peg.
- P19 marks the end of the peg, at its tip.

The two points must lie on the same (inferior) side of the component so that the line P18–P19 follows the peg axis. The order matters: P18 first, then P19.

P20 equivalent to **P18**

[New] P21 - the superior glenoid tubercule

According to https://www.jshoulderelbow.org/article/S1058-2746(18)30117-4/fulltext

[New] P22 - **the inferior glenoid tubercule.**

It will be used to measure the overhang.